What's Privacy Good for? Measuring Privacy as a Shield from Harms due to AI Inference of Personal Data

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Abstract

We propose a harm-centric conceptualization of privacy and operationalize it in the context of using artificial intelligence (AI) in education and employment. In an online study (N=400), US college and university students reported their perceptions of 14 harms (e.g., manipulation) when AI infers personal data (e.g., demographics and personality traits) and use it in decision-making.

We demonstrate that our approach can reliably and consistently measure privacy, sidesteps many limitations in existing frameworks, and captures harms from modern technology that would remain undetected by other frameworks. We surface nuanced perceptions of harms, both across the contexts and participants' demographic factors. Based on these results, we discuss how privacy can be improved equitably and inclusively. This research extends privacy theory and provides practical guidance to improve privacy in various technology use domains.

CCS Concepts

• **Security and privacy** → **Social aspects of security and privacy**; **Privacy protections**.

Keywords

Privacy, Artifical Intelligence, Inference, Privacy harms, Privacy framework

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1 Introduction

Privacy perceptions strongly influence privacy behaviors [62]. Their precise measurement is crucial to understanding why people adopt technology that may impact their privacy and designing privacy-enhancing mechanisms that cater to people's practical needs. Consequently, measuring how people perceive privacy and related constructs (e.g., concerns about privacy risks) have been among the central goals of human-centered privacy research.

Earlier efforts to measure privacy centered around the concept of concealment of private data, where privacy is *violated* when concealed data is disclosed [123]. However, categorizing data as private (versus public) is arbitrary and malleable [107], their disclosure is often a necessity or forced [63, 100], and they can be *inferred* using data considered non-private. Research on defining and measuring related (or proxy) concepts, such as privacy concerns [25], remained unsatisfactory [21]; these conceptualizations led to (now debunked [63, 105]) theories like the privacy paradox to explain concern-behavior mismatches. Likewise, a related conceptualization of privacy as *control over personal data* led to measures like notice and consent, which ultimately degraded privacy [106].

A different perspective, privacy as contextual integrity (CI) [79], focuses on data flows (rather than defining data as private or public) and declares them privacy-violating if misaligned with established norms in a given context. CI overcomes many of the limitations mentioned above, better explains people's privacy-related attitudes, and has gained wide popularity in the last two decades within the privacy research community [65]. CI, however, only concerns semantically meaningful data (e.g., age); it excludes fine-grained data (e.g., clicks on a website) that, when aggregated, can help infer semantically meaningful information. This exclusion renders CI incapable of determining privacy violations for a large number of digital services based on AI (artificial intelligence) that constantly perform such inferences. Moreover, CI's reliance on norms prevents it from recognizing novel privacy violations from established data-disclosing practices (see §2.1.2).

In this paper, **we conceptualize privacy as *a shield against perceived harms arising from the use of personal data in a given context***. The fact that privacy is much more than data protection [7], and its role in achieving other human values—including freedom, protection from manipulation, and identity building—has long been recognized [95]. A lack of privacy, then, may lead to harms like being discriminated or manipulated; such harm-centric framings are frequently used in court cases [7, 14] (about privacy violations) and legal scholarship [107]. This conceptualization avoids many limitations of the other formulations: it is oblivious to whether the data are private or sensitive, or accessed directly versus inferred from other data. Crucially, this framing may better explain and predict privacy-related behaviors, as privacy risks largely depend upon how data subjects perceive the negative effects of concrete privacy harms [9]. Finally, defining privacy as a set of harms that can be ranked based on (perceived) severity will allow developing technical and legal mechanisms to prevent specific harms rather than aiming to eliminate data disclosure, which is close to impossible in practice [63, 100].

Despite the many benefits, a harm-centered notion of privacy has not been formalized in human-centric privacy research, and its

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usefulness in measuring perceived privacy has not been evaluated. To fill these gaps, we study this notion of privacy in the case of using AI in two contexts: predicting if a student will drop out of a course or evaluating job candidates. Both tasks are increasingly being automated with AI models that make these decisions using a vast array of personal data [1, 23, 27, 42, 48, 53, 64].

In this study, we surveyed US college or university students (N=400). We focused on this group as they are current or future job seekers, and thus the population likely to have interacted with AI and be impacted the most by automation—the "experiential experts" [125]. During the study, participants indicated whether they (dis)agree that the use of personal data will cause harm in either the education or the employment context (between-subject design). The study included six classes of personal data (e.g., demographics, personality traits, and emotional state) and 14 types of harms (e.g., manipulation, bias, and stereotyping), selected based on a comprehensive literature review (see §3). Using data from this study, we investigate the following research questions:

RQ1: Can we reliably capture the notion of privacy through the lens of harms from data use?

RQ2: How do people perceive harms and do their perceptions vary across data types, contexts, and data subjects' demographic factors?

To answer **RQ1**, we first investigated the reliability and validity of measuring privacy through harm statements. Our extensive statistical analyses demonstrated high consistency and reliability, and that perceptions of all harms stemmed from one latent construct (§4.2). Moreover, we demonstrate that this latent construct, which we label as ***privacy harm perception***, was invariant across contexts, data types, and population segments (§4.3). Jointly, these results suggest that the harm-centric notion can be used in other context and populations to understand privacy.

For **RQ2**, our analyses revealed interesting and nuanced patterns in participants' conceptualization of privacy harms. For example, participants anticipated harms (e.g., being stereotyped, discriminated, or manipulated) from using demographic or personality trait data, **even when they did not think such data was inherently private**. These results lend support for the harm-centric understanding of privacy; it is not enough to just consider if data is private or public without looking at what harms might arise from its use. Comparative analyses across participants' demographics uncovered complex variations of perceived harms. For example, female and older participants consistently expressed more concerns about different harms compared to male and younger participants, respectively; however, the types of harms they were concerned about varied across data types and contexts (education versus employment). Comparison results were mixed across the races: white participants expressed more concerns regarding certain harms (e.g., privacy) compared to non-white participants, while trends were opposite for other harms (e.g., discrimination). We explain how such perceptions may have been shaped by social structures and lived experiences.

These findings not only provide empirical evidence of harm perceptions from AI tools but also underscore the benefits of the harm-centered framework. For example, this framing offers a more nuanced picture of privacy violations (compared to treating privacy as a monolithic concept [103]), and detects privacy violations that would go undetected under other frameworks (see results and discussions in §4). This approach identifies population groups that are the most vulnerable to specific harms and facilitates developing targeted prevention measures when a complete ban on data disclosure is infeasible or undesirable. Thus, this framework advances inclusive and equitable privacy. Moreover, our framing and methodology can be a valuable tool to quantify privacy perception, use that as a basis to predict behaviors, and design privacy-enhancing mechanisms that are practical and useful. In sum, this paper makes the following contributions:

(1) We formalize a harm-centric framework of privacy and operationalize it to measure perceived privacy in two widespread technology use contexts.
(2) We empirically demonstrate the reliability and consistency of measuring privacy through perceived harms. Thus, we extend the literature on privacy theory and frameworks. We discuss how a harm-centric framing can be benefit research and practice of discovering population segments most vulnerable to a privacy harm, privacy risk assessment, and predicting people's privacy decisions. We further demonstrate how this framework can facilitate equitable privacy.
(3) Besides theoretical contributions, this study reveals what harms people anticipate from AI in education and employment domains and offers how they might be prevented in practice.

We believe this research will directly help mitigate privacy harms caused by AI-based decision-making in education and employment. We also hope our framework will guide future research in understanding privacy in other contexts.

2 Background and literature review

2.1 Privacy frameworks

2.1.1 Privacy as concealment of (or control over) data. The bulk of privacy research conceptualizes privacy as the concealment of data that is considered 'private' or 'sensitive' (see chapter 2 of [62] for a review); privacy is violated when such data is leaked. This conceptualization has many limitations. The boundary of 'private' and 'non-private' data is arbitrary and malleable, and data considered non-private can be used to infer 'private' data [107]. The use of this framing to investigate people's mental models and concerns about private data leaks has unveiled phenomena like the so-called *privacy paradox*—when people express concerns about data leaks but behave in ways that leak private data—which has been debunked in latter works [63, 105]. The focus on protecting private data is partly to blame for this paradox: in an increasingly networked world, where digital tools mediated much of the social and professional activities, people, even those with the strongest privacy concerns, have no choice but to use services that require information disclosure [38, 86]. Opting out of digital tools use is tantamount to opting out of society [94], and thus nondisclosure of private data is not viable [47].

A related conceptualization of privacy is control over personal data [62], which is widely used in privacy laws and scholarship [108].

Yet, exercising this control is difficult, if not impossible, due to the same reasons mentioned above. On top of that, the current practice of providing control to data subjects is typically through notice and consent, which, in reality, exacerbates privacy violations by creating unrealistic cognitive burden, desensitizing people to such notices, or being manipulative [106]. Even if people were able to meaningfully exercise control over their data, they could do it only when the data was still undisclosed; this framing falls apart when one party (e.g., a data broker) gets access to the data, and the data subject cannot exercise control over the future transfer or use of that data. Finally, data obtained through consent can be subsequently used in a way that harms data subjects [106]; e.g., technology providers often sell data to third parties or data brokers, increasing numerous risks for the data subjects [45, 97], such as identity theft and harassment. Thus, the current conceptualizations of privacy fail to provide a useful framework for capturing harms from privacy violations [7].

2.1.2 Privacy as Contextual Integrity. Nissenbaum's theory of privacy as contextual integrity (CI), which views privacy as the flow of data according to established norms in a given context [80], has become the dominant framework to understand and measure privacy. According to Nissenbaum, "what people care most about is not simply restricting the flow of information but ensuring that it flows appropriately." [80]. This formulation side steps problems like arbitrary classifications of private and public data. Contextualizing data flows allows precise estimation of privacy based on what data subjects consider as privacy violations (or not) as opposed to more general and sometimes vague notion of privacy concerns (which may mean nothing without a context[1]). CI has been used in numerous studies to understand privacy (both normative and descriptive senses) as well as to predict people's behaviors related to technology use [65].

Yet, CI's one of the major limitations is, as Nissenbaum explained, that it only considers flows of semantically meaningful data (such as a person's age) and excludes finer grained data (such as online browsing behaviors) or information that can be inferred from other data (e.g., inferring someone's age from a photo posted online) [81]. Thus, CI cannot explain privacy violations in a significant portion of the data flows on today's digital platforms, which are driven by AI tools that predict or infer information based on user data.

Another, more fundamental limitation of CI is that its reliance on norms to define appropriate data flows precludes it from detecting privacy violations resulting from established behaviors (that have become the norm). For example, if posting videos online has become "normal" in a society, then CI cannot identify any privacy violations from such practices, even as technological advancements create new privacy harms (such as using recently developed generative models to create deep fake pornography from previously posted videos [114]). As Van de Poel argues, "the inherent conservatism of CI makes it possible to justify problematic technologies that follow established norms" [115].

[1]A commonly used example is, medical data flowing to a doctor may not be considered a violation of privacy, but if the same data flows to a drug marketing company, then it might be considered a privacy violation. But if people are asked about the data flow without providing a context, their response could be arbitrary and noisy

2.2 Harm-centered framing of privacy

Instead of asking 'what data needs to remain concealed' or 'which data flows need to be prevented' to protect privacy, we ask '**what harms can result from the use of personal data**' and conceptualize privacy as a shield against those harms. This formulation builds on various theoretical and applied research in privacy, law, and social science. Social sciences and legal scholarship have long recognized that privacy is much more than data protection [7] and is intimately related to many other human values, including freedom, protection from manipulation, and identity building [95]. Increasingly, researchers have begun to identify different types of information-based harms as privacy harms [7], and established privacy's significant role to prevent harms that arise from personal data use, such as discrimination and algorithmic manipulation.

In his recent work, prominent privacy scholar Daniel Solove persuasively argued to focus on harms from data use, rather than on the data itself [107]. Solove gives the example of a religious leader who might very much want people to know his role (to maximize impact) but would not want this information to be used to discriminate against him. Note that the harm arises due to the way information was used (i.e., to discriminate), rather than merely disclosing that information. Effective privacy protection would prevent the harmful use of personal data or mitigate resulting negative effects, rather than keeping the data secret. As Solove argued elsewhere [104]: "A theory of privacy should focus on the problems that create a desire for privacy." In another work, Citron and Solove pointed out the inconsistencies in court laws in recognizing privacy harms [19] and provided a typology of privacy harms that court cases can use.

A harm-centric conceptualization of privacy sidesteps the issues with defining data as private (or public) and is oblivious to how data gets disclosed. Rather, it makes a case against using data in ways that harm data subjects, even if consent was obtained or the data is already public. As opposed to norms, which by definition are dominated by the majority, the harm-centric framing allows uncovering and addressing negative consequences faced by population groups of any size, composition, and characteristics. Importantly, this framing can potentially explain and predict people's behaviors better than other frameworks, since harms are the realization of risks, and past research has shown that concrete risk understanding aids decision-making better than abstract privacy concerns [30]. Additionally, past studies reported that people are concerned about how data can be used in harmful ways rather than about the act of merely providing the data [51], and that people behave in ways to avoid negative consequences from technology use, even when they do not actively think about privacy [20]. Thus, a harm-centric framing can be useful in encouraging privacy-conscious behaviors, which has been a major goal in human-centric privacy research. Finally, focusing on concrete harms can provide clarity about the nature of privacy protections afforded by computational mechanisms. For example, differential privacy, one of the most popular techniques, provides a quantifiable privacy guarantee. Thus, if an AI model is trained in a differentially private manner, any information extracted from that model cannot be linked back to individuals whose data was used in the training. Such a model, however, still may harm other people's privacy; e.g., if, after deployment, an

AI-based hiring agent learns secrets about job candidates. These issues will be characterized as privacy harms under the proposed framework.

Despite compelling benefits, measuring privacy in terms of harms in a given context has not been formally explored in the literature. Several past works, however, empirically identified privacy harms. For example, Karwatzki *et al.*reported seven harms (e.g., physical, social, and psychological) that individuals perceive if their privacy is invaded [56]. Jacabi *et al.*expanded this research by eliciting 91 privacy harms through focus group interviews and workshops. Later, they employed experts to categorize those harms to create a high-level taxonomy [52]. Wu *et al.*conducted a survey of privacy harms from online behavioral advertising and identified four key harms: psychological distress, loss of autonomy, constriction of user behavior, and algorithmic marginalization and traumatization [122]. Analyzing documented privacy incidents involving AI tools, Lee *et al.*identified 12 high-level privacy risks [68]. More related to our context of investigation, Pyle and Andalibi uncovered US college applicants' perception of harms (e.g., hindering diversity) from algorithmic decision-making in the context of admission [92]; their another work investigates privacy harms perceived by job seekers due to AI-enabled interviewing platforms [93]. In the education domain, several prior studies investigated stakeholder privacy concerns due to rampant technology use [41, 124]. Our work has been heavily informed by these prior arts, but is different in that we focus on measuring privacy through perceived harms. Finally, Solove's seminal work created a taxonomy of privacy [103] that includes 16 activities (e.g., surveillance) that lead to privacy harms, whereas we aim to surface privacy harms from a given activity (e.g., AI using inferred personal data).

2.2.1 Harm-centric privacy in the Government and Industry. Outside academia, government agencies and corporations have started to adapt risk-based (the product of the impact of a privacy harm and its likelihood) frameworks to privacy, often conceptualized as the negative effects perceived by data subjects [9]. For example, NIST (National Institute of Standards and Technology) has begun to frame privacy issues in terms of risk, drawing parallels to security violations as loss of confidentiality, integrity, and availability of a system [10]. Article 25 in the General Data Protection Regulation specifically refers to assessing the severity of privacy risks [3], which can be quantified based on the severity of different privacy harms caused by privacy threats. Recently, MITRE Corporation has released a comprehensive frameworks for privacy threat modeling called PANOPTIC [57]. It defines privacy attacks based on perceived harms: any action or inaction that could be perceived by affected individuals to cause a privacy harm [57]. These examples demonstrate the necessity and practical usefulness of focusing on privacy harms; however, a formal treatment to *measuring privacy as a shield against specific harms in a given context* has yet to be done. Such a formalization is essential to make theoretical advancement in understanding privacy itself and to complement existing privacy frameworks.

2.2.2 Procedural justice and privacy harms. We use the lens of procedural justice theory [70] to formalize harm-centric privacy. Procedural justice examines if the process of a decision-making task is just and fair toward people who might be impacted by that decision [70]. Such investigations are necessarily empirical and bottom-up; thus, they require surfacing the perceptions of the affected population (e.g., see [31, 35, 77]. To mitigate harms from algorithmic predictions, legal scholars Crawford and Schultz proposed the concept of 'the right to procedural data due process' [22]—that is, individuals must be able to demand that predictive models be built following a just process to minimize negative impacts on them. This paper operationalizes this idea in the contexts of automated decision-making in education and employment through an empirical study.

2.3 Use of AI in education and employment

We briefly review the literature on AI use and associated privacy concerns in education and employment, two contexts this study focuses on.

2.3.1 AI-based prediction tasks. AI and machine learning-based models have become ubiquitous in both sectors. Some of the most common predictive tasks in education include course performance prediction [119], the probability of dropping out from a course [23], students' attention and behavior prediction [58], as well as detecting students' cheating behaviors during remote exams [41, 113, 124]. In employment, AI models evaluate potential candidates' job performance [1], likelihood of a candidate accepting a job offer[29], and predicting employee turnover and retention[91]. Additionally, AI systems are being employed to screen resumes, conduct initial interviews, and even make hiring recommendations [66].

2.3.2 Use of personal data and resulting privacy harms. Various types and modalities of personal data have been used to train predictive models in education, including demographics and socio-economic status, behavioral data (login patterns, time spent on tasks, interaction with resources), and even mobility and activity patterns. Likewise, AI in employment uses resume and online portfolio, assessment scores (cognitive ability tests, job knowledge tests), interview data (verbal responses, facial expressions, voice patterns), and external information (social media profiles, public records) [6, 16].

Privacy researchers and activists have raised numerous concerns about AI models using such data [42, 75, 101], and many laws prohibit using many of these data items as decision criteria (e.g., gender and race for college admission or hiring employees). However, recent research has shown that legal measures to prevent direct use of these data items do not eliminate privacy harms. For example, many behavioral data encode demographics, socio-economic, physiological, and psychological characteristics [42, 44, 109]; this information can be inferred by re-purposing AI models (originally trained to predict, e.g., students' course performance) or extracted from those models' internal representation of input data [42, 44]. Moreover, even in the absence of an active adversary, the models may implicitly rely on such data to make decisions, resulting in biased, exclusionary, and stereotyped decisions. Concerningly, many AI systems deployed in the real world are already exhibiting such behaviors [64, 84, 121].

Unfortunately, the current conceptualizations of privacy fall short in characterizing such incidents as 'violations of privacy' [7], as data is already out of control (public) or "voluntarily" provided by the data subjects. However, the role of privacy, as broadly understood, includes providing freedom from manipulation, discrimination, and marginalization, among other things, and such framing would rightly categorize these incidents and privacy harms, of which this paper provides empirical evidence.

3 Methodology

3.1 Study design

3.1.1 Study context selection. An investigation of harm-centric conceptualization of privacy needs to be context-dependent, because the same data may lead to different privacy harms when used in different contexts. For example, revealing the disability status to an academic advisor will affect an individual differently than revealing it to a future colleague. To contextualize privacy harms, this study focuses on the harms from AI use in two domains: education (AI predicting the probability of students dropping out of courses) and employment (AI recommending job candidates). Their selection was motivated by, as § 2.3 details, AI's ubiquity in these contexts and their use in making high-stakes decisions, and importantly, the possibility of privacy harms without any breach of 'private data' or an active adversary.

3.1.2 Target population and study method. To be comprehensive and inclusive, such investigations must also be bottom-up: gathering empirical evidence of harms perceived by people whose personal data will be used for these decision-making [52, 73, 125]. Ideally, they are also the *experiential experts*: people living the experience or those closely associated with someone living the experience [125]. Historically, the legal domain has identified privacy harms deductively (by basing on fundamental human rights violations); but recently scholars have raised critical questions about the authority, legitimacy, and inclusiveness when identifying new privacy harms this way [7]. To avoid such pitfalls, scholars recommend surveying the affected population [7]. Recent regulatory measures also stress the importance of including the voice of data subjects; e.g., Article 35(9) of the EU General Data Protection Regulation states that "Where appropriate, the controller shall seek the views of data subjects or their representatives on the intended processing, without prejudice to the protection of commercial or public interests or the security of processing operations." [3]. To align with these guidelines, we recruit study participants from the college and university student body, who are also current or future job seekers, and thus likely have experience with AI-based decision-making in those contexts (i.e., experiential experts).

With these design decisions, we conducted an online study to understand how students perceive the harms of using personal data by AI models to make decisions in education and employment.

3.2 Selection of personal data types

We focus on six types of personal data as shown below; their selection was guided by the following criteria: (1) data commonly used by AI systems in education and employment contexts, (2) data that are expected to generate a wide range of perceived harm level (i.e., high variance), and (3) data that may not be directly collected but can be inferred from other data.

(1) **Demographics**: The use of demographic data, such as age, gender, and socio-economic variables, as predictors in AI models has been extensively studied in the context of privacy, ethics, and fairness of algorithmic systems [61, 87]. While their direct inclusion as a decision criterion may be unlawful in many regions, such information may be encoded in other data (such as behavioral data [42, 109]), and extracted[2] by those who implement or deploy AI models. Further, these encoding may induce discriminatory behaviors in AI models [121].

(2) **Personality traits**: Personality traits, such as extraversion and agreeableness [34], have been extensively studied as predictors of academic performance [39] and employability [112]. Personality traits are reflected in other data, such as resumes and online profiles [54, 116], as well as social media activity and behaviors during job interviews [32, 54]. Research has demonstrated AI models' ability to predict traits from resumes and online profiles [6, 49], and many platforms explicitly use personality scores computed from such data to make hiring decisions [6].

(3) **Emotional states**: Multimodal data, including writing samples and recorded videos during online learning, have long been used to determine students' emotional states and use them for purposes like determining engagement with course materials and predicting course performances [26, 98]. In the hiring context, analyzing video interviews to understand affective states and other related constructs (e.g., confidence and communication styles) has become commonplace [93]. Such inferences raise concerns about privacy, ethics, and potential misinterpretation of emotional data [96].

(4) **Motivation:** Motivation is frequently used as a predictor in both educational and employment contexts[71]. Motivation is generally considered volitional and within one's control, and its use in education or hiring contexts may be more acceptable compared to other data types. Motivation, however, is a complex construct that may manifest through diverse behaviors [85], and its automated measurement on a limited sample and form of behavioral data may be inaccurate or unreliable. Furthermore, other characteristics may be correlated with this construct (such as race), leading to biased estimations by AI models [55, 121].

(5) **Creativity and problem solving:** Like motivation, creativity and problem-solving abilities are increasingly used in both educational assessment and hiring decisions [12]. While these attributes may have legitimate uses in both contexts, measuring them through scattered behavioral data or interaction patterns without scientific validation raises concerns about consistency, reliability, and potential bias.

(6) **Physical or cognitive impairment:** Information about physical or cognitive impairments may be directly disclosed

[2] we use this term loosely to indicate various actions, such as re-purposing a pre-trained model to predict people's demographics or prompting large language models to make it reveal memorized training data.

by individuals seeking accommodations or indirectly inferred through multi-modal data. For example, current machine learning models can learn about speech impairments, limited vision, and learning or cognitive disabilities from audio, image, and video data [17, 78, 120]. Leaking such knowledge may result in harassment and safety issues for individuals, and (implicit or explicit) use of disability status in decision-making will likely yield discrimination [33, 83].

3.2.1 Selection of privacy harms. Our selection of privacy harms builds on existing harm taxonomies [4, 52, 56] and literature on human-centered privacy [11, 42, 68, 69, 92, 101, 110], law [13, 19, 118] and ethics [8, 74], procedural justice and fairness [22, 35, 70]. Our selection of harms was constrained by the contexts we focused on; that is, what data AI-based tools use when predicting dropout from a course or recommend job candidates, what inferences could be made from that data, and, crucially, how that inferred data could harm data subjects. As such, we excluded many harms that were part of existing literature: e.g., physical [4, 19, 52], resource-related [52], prosecution-related [52, 56], and political harms [4]. Since we focused on harms for the data subjects (i.e., individual), we also excluded societal or collective harms [52], environmental harms [4], political harms [4], and many harms that were categorized under business (e.g., Monopolisation [4]). We also found different namings and conceptual overlaps among multiple harms: e.g., health impairment (due to unfavorable uses of data [50]) and Discrimination [19] were covered by *Bias* in our taxonomy. Finally, we included several harms that may arise because of how AI (and machine learning) models work, and may not be relevant to general data use. For example, the training process can optimize AI models to favor profit for the owners over data subjects rights [74] and training data may contain irrelevant predictors (e.g., gender in predicting course performance [42]) that may degrade model reliability and consistency. Our final set contains 14 harms, including both material (objective) and psychological (subjective) harms as defined by Calo [13]. We describe them below:

(1) **Invasion of privacy:** Data is considered inherently private, and just their unwarranted exposure can lead to psychological discomfort [4, 19, 52]. Constant or unexpected surveillance can further interrupt one's regular life and create chilling effects, thwarting self-expression and autonomy [19].
(2) **Bias:** It involves discriminating against certain groups of people based on gender, race, sexual orientation, age, disability, or other characteristics and affiliations [4, 19, 70, 92].
(3) **Loss of autonomy:** This harm, taken from Abercrombie *et al.*, arises as AI inferring personal information and using it for decision-making can hamper an individual's control over their data and how they wish to be seen or judged by others [4]. It also relates to 'coercion' as conceptualized by Citron and Solove [19], since students or job applicants have no choice but to disclose personal data.
(4) **Stereotyping** Derogatory or otherwise harmful generalization of individuals on demographic or behavioral properties and was categorized under societal harms by Abercrombie *et al.* [4]. In our contexts, it may arise due to the knowledge of (through inference) race, gender, or other social categories.
(5) **Violation of ethics:** Subjecting individuals to algorithmic predictions solely for the service provider's benefits and efficiency can be unethical and an arbitrary exertion of power [5, 22, 74]. This also relates to Economic harms (loss of opportunity) to data subjects [19] since solely focusing on provider's efficiency may result in sub-optimal results for data subjects, and more so for minority groups, since the measurements and optimizations are often defined in ways that correlate with systematically disadvantaged groups in the society [74].
(6) **Defamation:** It involves an individual's loss of reputation within their social or professional communities and has been incorporated in multiple harm taxonomies [4, 19].
(7) **Harassment:** Inferred data can be used for online/in-person harassment [4, 11]. This harm is closely related to the psychological harms by Citron and Solove [19].
(8) **Manipulation:** Inferred data can be used to covertly alter user beliefs and behavior using nudging, dark patterns, and other opaque techniques [4, 19, 110].
(9) **Stress:** Citron and Solove ranked emotional distress among the most frequent harms caused by privacy violations, which includes various emotions like annoyance, frustration, and anxiety [19].
(10) **Inaccurate decision** The use of certain predictors can lead to inaccurate decisions [70, 101, 118]. In our context, it might result in biased outcomes, and economic harms [19] as people may lose and opportunity (e.g., employment, performance-based academic awards).
(11) **Lack of user control:** When people are judged over something that is not under their control and so they cannot change it to be judged differently, they might perceive the judgment process as unjust and harmful [8, 35, 70]. In our contexts, AI models regularly include predictors, such as gender and age [42], in decision-making where people can feel helpless to do anything to get a favorable result.
(12) **Inconsistency:** Leventhal, in his seminal work on procedural justice, stated that for a process to be just, it must be consistent across people and over time [70]. Relying on certain (inferred) predictors can lead to inconsistent decisions when the predictor's distribution changes; this phenomenon is widely known withing the machine learning community as distribution shift, which was found in almost all areas of AI use (e.g., [69]).
(13) **Lack of relevance:** When some data is used in a decision-making process, if that data is irrelevant to the outcome, its could be considered as unjustified and harmful to the data subject (e.g., by resulting in an unfavorable outcome). This harm, inspired by legal requirements that any admissible evidence be relevant to the case, was studied by in the criminal risk prediction context [35]. Irrelevant data use frequently appear in predictive modeling; e.g., in the education context, Hasan *et al.*showed that demographic data, while regularly used to predict student performance, do not have any causal relevance to that outcome [42].

(14) **Wrong inference:** The predictor (e.g., age) may not itself be reliably inferred from raw data. This harm may arise due to the inherent limitations of machine learning models, which are unlikely to be always correct. Such wrong inference was explored in the case of criminal risk prediction [35], and violates people's right to reasonable inference [118]. Since such inferences are implicit or internal to the AI models, they goes undetected; consequently, data subjects are not informed or provided any chance to correct them [19].

3.3 Data collection

Participant recruitment. After obtaining approval from our institutional ethics board, we implemented this study on Qualtrics and recruited participants on Prolific [2]. Prolific provides filters to recruit participants with specific properties (such as location, and current employment status). Using these filters, we advertised our study to the target population: people who are living in the US and are currently enrolled at a US college or university (including those at community colleges). Upon acceptance of our invitation, each participant answered questions related to either the education or the employment context (randomly selected).

Survey flow and instructions. In the survey, after consenting, participants were briefed about the use of AI in education (or employment, depending on the randomly assigned condition). Following that, we explained how an AI model can learn information (through inferences) about a person (e.g., demographics) using various data sources (e.g., images found online) and use that in predicting dropout risk (or fit for a job). Participants then indicated their (dis)agreement with statements about such inferences being harmful (e.g., "Use of personality traits to decide dropout risk can lead to biased decisions") using a five-point Likert scale ("Strongly Disagree" to "Strongly Agree," coded from -2 to 2 during analysis). To ensure data quality, we included one question hidden from human eyes but visible to an automated survey parser algorithm. Additionally, we included one qualitative question at the end and manually reviewed the answers to identify gibberish responses. The instructions and full questionnaire are provided in the Appendix (§ A.1).

Note that, explaining how AI-based tools work was necessary for the participants to understand the context and the questions. But such explanations could bias participants or lead them to a certain response. To avoid this, we explained how AI works, mentioned rationales for their use (e.g., increased efficiency), but did not mention any negative impacts. We then asked about their perceptions of harms; the question format closely followed prior similar studies [36, 117]. Note also that our goal was not to measure people's awareness of possible harms, which could have been inflated by the mention of the word "harm." Rather, we aimed to measure how much harm they think certain data uses may cause; estimating this requires some understanding of how the data will be used. Finally, the premise of this research is that eliminating all harms by preventing data disclosures is infeasible; rather, the goal should be minimizing harms (akin to risk minimization [57]). Thus, their relative rank is more important than absolute scores; consequently, the effect of bias in the measurement, if any, is negligible.

Pilot study. We first recruited 20 participants and used their responses to conduct a power analysis (power level 0.8), which indicated that we need at least 392 participants for a significance test at a false positive level of 0.05. We also manually reviewed the responses to a feedback question that asked if the instructions were clear and if participants understood the questions and harm categories. No major issues with the survey were reported; we did minor revisions (e.g., rephrasing) of the instructions based on the feedback.

Main study. Next, we collected data from 430 more participants, but discarded 30 responses because of the high likelihood of AI-generated responses (because they answered the hidden question, N=4) or inattentive responses (detected by manual review of qualitative responses, N=26). The median time for study completion was 16 minutes and 40 seconds. All participants (except the 4 AI bots) were compensated with $5 for completing the study.

Limitations. Our study may suffer from sample bias as we collected data through an online platform. Additional study may be needed to investigate if our results generalize to the whole US population of college and university students. We limited our investigation to six attributes (data types) to keep the study manageable in length. Another important limitation is that we investigated privacy harms in contexts that required the use of personal data. However, mere surveillance can cause privacy harms (e.g., creating chilling effects and inhibiting the freedom of expression [103] even if no data is collected or used. We leave such investigations to future research. Finally, since privacy is inherently contextual [80], we did not aim for generalization to other contexts. This was a deliberate design choice: the premise of the harm-centered notion of privacy is that using data in different contexts may lead to different harms. Finally, we acknowledge the possibility of common method bias [15] being present in our analysis, which is a persistent issue in human subject studies [89].

4 Results

4.1 Participants

Among the 400 respondents, 45.3% (N = 173) self-identified as males, 52.1% (N = 199) as females, 2.4% (N = 9) as non-binary, and the rest preferred to self-describe (N = 1). Participants' age distribution showed the following: between 18-24(42.2%, N = 161), between 25-34 (38.5%, N = 147), between 35-44 (13.6%, N = 52), between 45-54 (4.7%, N = 18), and 55-64 (N = 4). Participants' racial distribution showed the following: White (58.2%, N = 219), Black or African American (28.5%, N = 107), Asian (9.3%, N = 35), American Indian or Alaska Native (2.7%, N = 10), and Native Hawaiian or Other Pacific Islander (N = 5). A majority of participants indicated that they were Undergraduate/Bachelor (53.3%, N = 201), followed by Masters (19.4%, N = 73), Technical/Community College (16.5%, N = 62), Doctoral (9%, N = 34), and others (N = 7). Participants were classified based on their field of study: STEM (47%, N=189), and Non-STEM(53%, N=211). We had an equal split across the experimental conditions, that is, each of the education and employment contexts had responses from 200 participants.

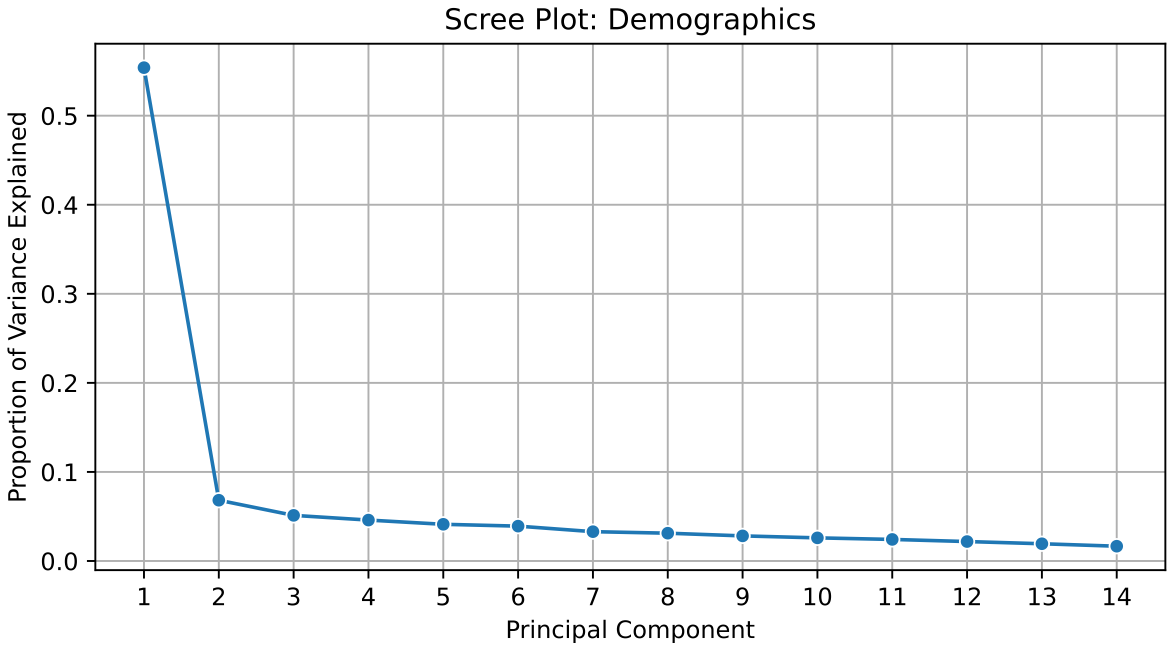


Figure 1: Scree plot showing how much variance was explained by principle components.

4.2 Measurement reliability and validity

We computed Cronbach's alpha, which indicates internal consistency of a set of items [111], in this case, the 14 harm statements. Table 1 shows the alpha values for the education context (the first row), the minimum alpha being 0.93, the items indicate high reliability [111]. The subsequent rows in Table 1 show alpha scores when one item was removed at a time. Removing any item decreased the score, indicating their complementary roles. Each item also correlated highly with the mean of the other items (i.e., item-total correlation, shown within parentheses). We obtained similar results for the employment context (see Table 5 in the appendix).

Next, we conducted a factor analysis to assess the dimensionality of the latent construct that the 14 items measured. We varied the number of dimensions from 1 to 14, created scree plots, and used the elbow method to select the final number of dimensions [28]. The scree plot for the Demographics data type in the education context is shown in Figure 1; the variance explained sharply declines after the first factor, indicating a single latent dimension. All of the remaining 11 plots showed the same pattern; we omit them to save space. The total variance explained ranged from 52% to 66% in the education context and 50% to 61% in the employment context. These results indicate that the single factor contains sufficient information about the items (50%–60% variance is considered a good fit in social science research [28]).

The above results indicate that all items jointly measure one underlying or latent construct, which we term as ***privacy harm perception***. Since we conceptualized privacy as a shield against various harms, the items indicate how much *privacy* one loses due to personal data use, or how much *privacy* they would gain by preventing that data use (protecting privacy). Thus, under this harm-centric notion, this construct indirectly measures privacy itself. We do not, however, claim that study participants consciously thought about privacy while answering the questions. Nor do we believe it is necessary, because our goal is to measure (and then prevent) harms from personal data use. Past research reported that people remain unconcerned about privacy when presented as a monolithic concept, but concrete examples of privacy breaches improved their risk comprehension and behaviors [40]. More, past research has also shown that people attempt to avoid privacy harms without consciously thinking about privacy [20]. The harm-centered notion of privacy was partly motivated by these findings, where the goal is to capture people's perceptions of harms from personal data use as the measure of privacy they need to prevent those harms.

Takeaways:
Statistical analysis demonstrated that all 14 items are internally consistent and they jointly measure a single latent construct: **privacy harm perception**. Thus, this approach to measure privacy through perceived harms is reliable and can be used in domains.

4.3 Measurement invariance analysis

The above results (§4.2) confirm that the harm items are consistent and reliable in jointly measuring one composite construct: perceived privacy harm. Here, we investigate whether that composite measure is consistent across subpopulations of the participants. That is, we test if the items are interpreted the same way by participants from

Table 1: Cronbach's Alpha (leave-one-out) and Item-Total Correlation (in parentheses) by Attribute for Education Context

Item Removed	Demographics	Personality	Motivation	Emotion	Creativity	Impairment
-	0.93	0.94	0.95	0.95	0.96	0.96
Unethical	0.91 (0.76)	0.93 (0.77)	0.94 (0.83)	0.94 (0.76)	0.95 (0.84)	0.96 (0.80)
Biased	0.92 (0.75)	0.93 (0.74)	0.94 (0.77)	0.94 (0.79)	0.95 (0.75)	0.96 (0.83)
Inaccurate	0.92 (0.74)	0.93 (0.74)	0.94 (0.76)	0.94 (0.78)	0.95 (0.78)	0.96 (0.80)
Privacy	0.93 (0.57)	0.93 (0.72)	0.94 (0.78)	0.95 (0.70)	0.95 (0.77)	0.96 (0.74)
Inconsistency	0.92 (0.72)	0.93 (0.71)	0.94 (0.75)	0.94 (0.79)	0.95 (0.80)	0.96 (0.81)
Freedom	0.92 (0.68)	0.93 (0.76)	0.94 (0.73)	0.94 (0.76)	0.95 (0.80)	0.96 (0.78)
Stereotyping	0.92 (0.72)	0.93 (0.71)	0.95 (0.70)	0.95 (0.67)	0.95 (0.77)	0.96 (0.80)
Reliable	0.92 (0.64)	0.93 (0.64)	0.94 (0.73)	0.94 (0.76)	0.95 (0.79)	0.96 (0.76)
Defamation	0.92 (0.71)	0.93 (0.73)	0.94 (0.74)	0.94 (0.73)	0.96 (0.71)	0.96 (0.75)
Harassment	0.92 (0.63)	0.94 (0.59)	0.95 (0.71)	0.95 (0.67)	0.95 (0.77)	0.96 (0.78)
Manipulation	0.92 (0.61)	0.93 (0.71)	0.94 (0.73)	0.94 (0.79)	0.95 (0.75)	0.96 (0.78)
Stress	0.93 (0.53)	0.93 (0.66)	0.95 (0.68)	0.95 (0.66)	0.96 (0.73)	0.96 (0.74)
Inference	0.92 (0.63)	0.93 (0.72)	0.95 (0.72)	0.94 (0.81)	0.95 (0.79)	0.96 (0.77)
Control	0.92 (0.64)	0.94 (0.55)	0.95 (0.62)	0.95 (0.59)	0.96 (0.70)	0.96 (0.74)

different demographic segments in different contexts. For this, we conducted a series of confirmatory factor analyses to check if the factor structure at the population level (i.e., a single factor) was retained across contexts (education and hiring) and demographic factors (age, gender, race, and education level). We followed the well-established procedure from Van de Schoot *et al.* [99] that comprises four steps, i.e., four confirmatory factor analysis (CFA) models, each more restrictive than the before. In all cases across contexts and demographics, the Configural CFA model showed sufficient model fit according to the widely-used critera from Hu and Bentler [46]: the comparative fit index (CFI) was $> .95$; Tucker-Lewis index (TLI) was $> .95$; and root mean square error of approximation (RMSEA) $< .06$. In all cases, when comparing the Configural CFA model with either the Metric invariance model or the Scalar invariance model, the results were non-significant ($\Delta\chi^2 > .05$), indicating that both metric and scalar invariance held. While these results are sufficient to compare the construct across groups [99], we also found that the strict invariance held for context and race.

Takeaways:
Invariance analysis confirmed that our harm measurement instrument was interpreted the same way across context, data types, and by participants across various demographic characteristics. Thus, our instrument can be used to measure privacy consistently within population segments and across situations, and the measured values can be compared to gain new insights.

4.4 Privacy harm perception and its variations across context, data, and demographics

How people perceive privacy is expected to depend on the data, use context, and personal factors like demographics. Additionally, we hypothesize that, for a specific harm, how people perceive to be vulnerable to that harm will vary based on their socioeconomic situation and lived experiences. We believe that surfacing these differential perceptions is crucial to identify and protect the most vulnerable in a given context.

Thus, here we examine how perceptions varied across data, context, and population segments; we do this both at the composite privacy harm perception level (i.e., the sum of individual harm scores for each data type) and the individual harm level. For the former, we built a mixed linear model where the composite harm score was used as the outcome, and context, data types, and demographics were used as predictors. This model was used to conduct analysis of variance (ANOVA), followed by post-hoc pair wise comparisons. For these comparisons, we created groups by dividing participants based on gender, age, race, education levels, and study discipline. Specifically, we compared males with females, participants 18–24 years old with those older than 24 years, 'white' with 'non-white' participants, and undergraduate students with postgraduate students (resulting group sizes are shown in Table 6 in the Appendix). To investigate group wise differences for individual harms, we conducted Mann-Whitney U tests with Bonferroni corrections for multiple tests. The following sections report results from these analyses.

4.4.1 Effects of context and data type. Table 2 shows results form ANOVA. The context of data use significantly affected privacy harm perceptions (Table 2). A post-hoc comparison between the two contexts showed that participants were more concerned ($t = 2.61, p < 0.009$) about privacy harms in the education context ($M = 5.2$, $SE = 0.74$) compared to the hiring context ($M = 2.5, SE = 0.34$). Table 2 also shows that the type of data significantly affected the privacy perception. Figure 2 shows that, among the six data types, demographics and disability status (physical/cognitive impairment) raised the highest concerns, while using motivation or creativity were not perceived to be harmful. However, privacy perception about the same data varied between contexts, as indicated by the significant *Context* and *Data type* interaction effect in Table 2 and visually depicted in Figure 2. We conducted post-hoc comparisons for each data type across contexts (Holm–Bonferroni adjustment was used to correct for multiple comparisons). We found that, participants expressed higher concerns ($t = 2.8, p < .05$) about personality traits in the education context ($M = 6.2, SE = 0.92$) than the job

Table 2: Main and interaction effects from Analysis of Variance

	Sum Sq	Mean Sq	NumDF	DenDF	F statistic	$Pr(>F)$
Data Type	42110	8422	5	1835	123.35	$< .0001^{***}$
Context	469	469	1	361	6.87	0.0091^{**}
Race Group	3	3	1	361	0.05	0.8219
Age Group	647	647	1	361	9.47	0.0022^{**}
Edu Level	4	4	1	361	0.06	0.8129
Gender	648	648	1	361	9.49	0.0022^{**}
Race Group : Age Group	112	112	1	361	1.65	0.2002
Age Group : Gender	4	4	1	361	0.06	0.8035
Context : Gender	350	350	1	361	5.12	0.0242^{*}
Context : Race Group	77	77	1	361	1.13	0.2876
Context : Age Group	6	6	1	361	0.09	0.7595
Data Type : Gender	461	92	5	1835	1.35	0.2402
Data Type : Race Group	883	177	5	1835	2.59	0.0243^{*}
Data Type : Age Group	555	111	5	1835	1.63	0.1497
Data Type : Context	2863	573	5	1835	8.39	$< .0001^{***}$

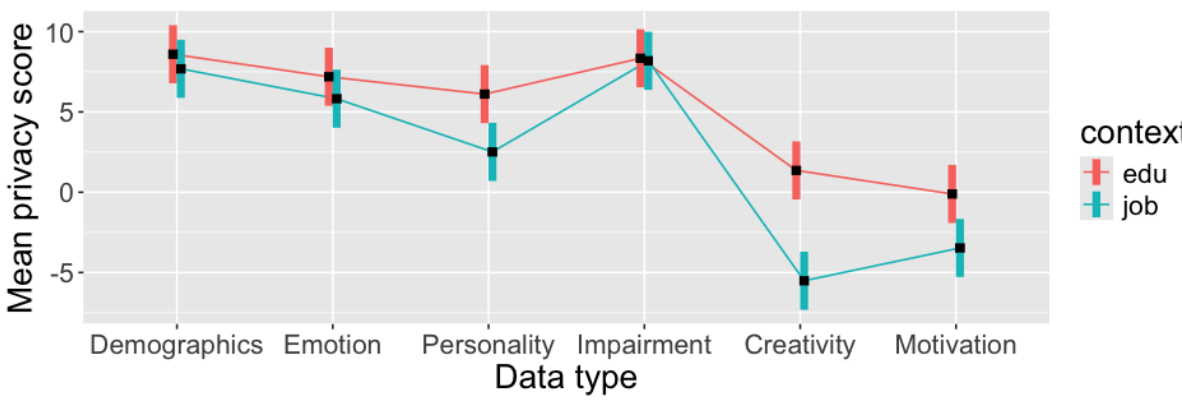


Figure 2: Mean and SE of perceived privacy harms across data types and contexts.

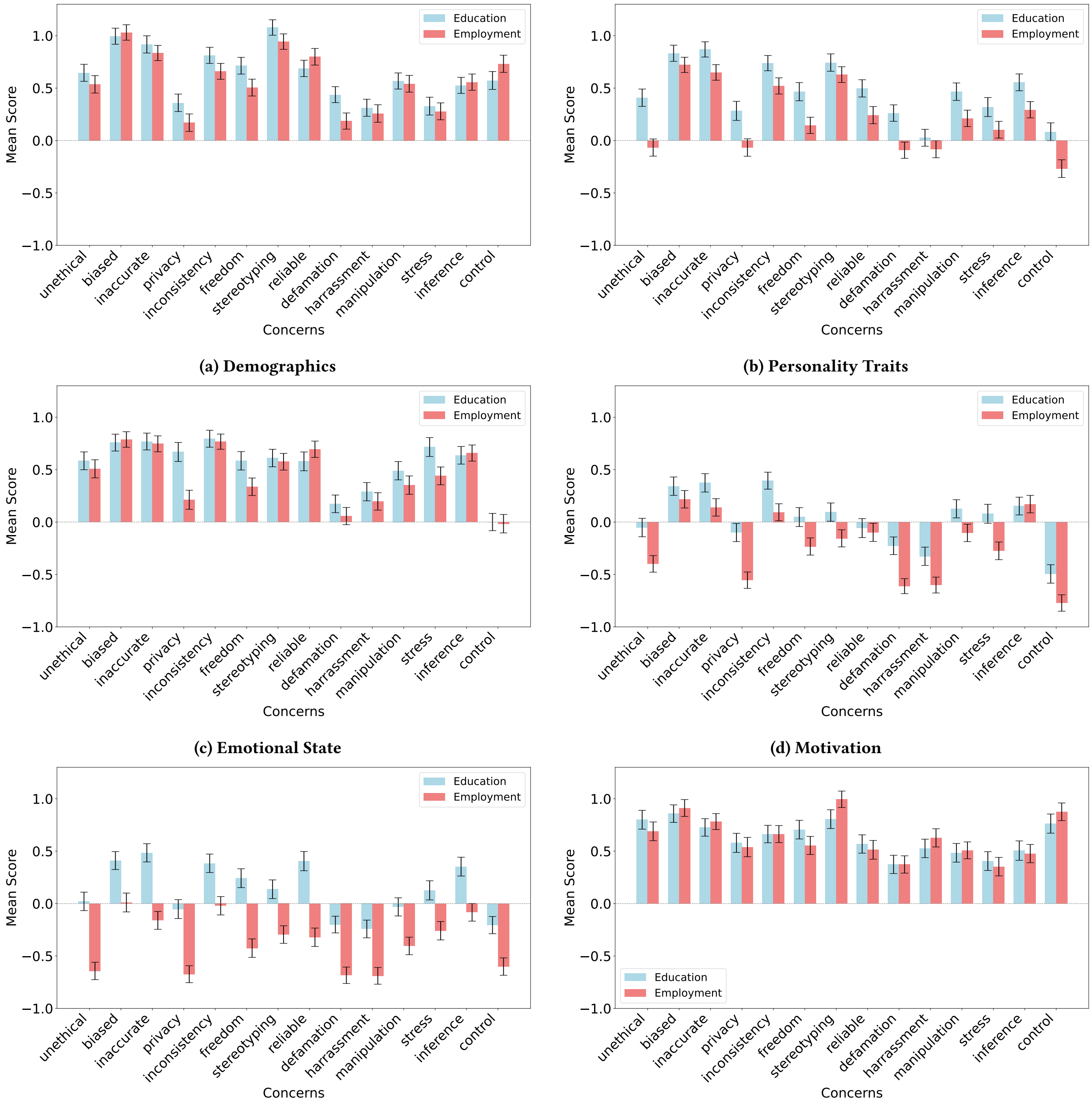
context ($M = 2.6, SE = 0.92$). In contrast, participants expressed concerns about motivation when used in education ($M = 0.1, SE = 0.92$) but not when used in the job ($M = -3.4, SE = 0.92$), and the difference was significant ($t = 2.6, p < .05$). Likewise, creativity also evoked concerns when used in education ($M = 1.4, SE = 0.92$) but not when used in the job context ($M = -5.5, SE = 0.92$, comparison: $t = 5.3, p < .0001$).

Figure 3 provides a more nuanced picture at the level of individual harm types: almost all harms can materialize when demographics, emotion, or disability status is used by AI models in both education and job context. Using personality traits was associated with bias, inaccurate prediction, and stereotyping in both contexts, but other harms were perceived only in the education context. Results between these two contexts diverged even more for creativity and motivation; using them was perceived to be harmful in the education context but not in the employment context.

We get an interesting insight from Figure 3: participants mostly disagreed that these data was inherently private (except for disabilities and emotional states), especially in the employment context. These results highlight that data considered non-private can lead to privacy harms, and that the same data can lead to different harms in different contexts.

Figure 7 in the Appendix shows separate plots for education and employment contexts, showing the number of participants in each of the response categories. They show how prevalent each of the perceived harms is among the participants. Importantly, they reveal the non-uniformity of harm perceptions among groups of participants, and that at least some people feel vulnerable to each of the harms, even if the general trend is negative (e.g., *Defamation* resulting from *Motivation*). Such a feeling of vulnerability may correlate with people's demographic and socio-economic factors, allowing us to identify the group(s) requiring the most protection against a certain harm.

4.4.2 Effects of participants' demographics. As Table 2 shows, participants' demographic factors (age and gender) predicted their harm perceptions. For age, as Figure 4 shows, older participants were more concerned ($t = 3, p < 0.002$) about privacy harms than younger participants ($M = 5.6, SE = 0.68$ for older versus



(a) Demographics

(b) Personality Traits

(c) Emotional State

(d) Motivation

(e) Creativity and Problem Solving

(f) Physical or Cognitive Impairment

Figure 3: Mean and SD of perceived harms in education and employment.

$M = 2.2$, $SE = 0.80$ for younger participants). Additionally, a post-hoc comparison across genders revealed that females were more concerned ($t = 3, p < 0.002$) about privacy harms than males ($M = 5.4$, $SE = 0.71$ for females versus $M = 2.3$, $SE = 0.75$ for males).

These demographic factors also interacted with context and data types (see Table 2). Figure 5 shows gender-context interactions, and Figure 6 shows data-race interactions. Post-hoc comparisons (with Holm-Bonferroni correction) showed that females expressed a higher level of concern ($t = 3.6, p < .0001$) when personal data

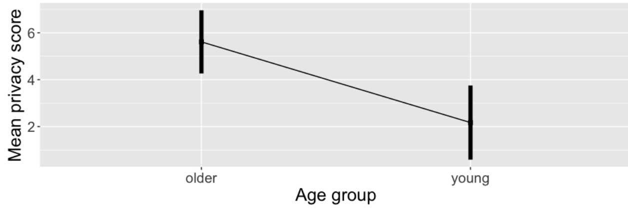


Figure 4: Mean and SE of perceived privacy harms across age groups.

was used in education context ($M = 7.9, SE = 1.01$) compared to job context ($M = 2.9, SE = 0.99$). But males did not differ across contexts ($p > .05$). While the interaction between race and data types was significant (Table 2), post-hoc comparisons did not find any significant differences between white and non-white participants.

Takeaways:

(1) The use of demographic, emotional state, or physical/cognitive disability status in either context was perceived as most concerning.
(2) Data not considered inherently private can still cause privacy harms.
(3) For the same data, perceived harms differed across contexts.
(4) Our approach surfaces harms that may impact only a minority of the population, thus advancing inclusive privacy.

4.5 Variations in perceptions at individual harm levels

To provide a more nuanced picture, we further investigated if participants differed in perceiving individual harms, even if the aggregated harm score did not differ. For this, we conducted group-wise comparisons (Mann-Whitney U tests) across the five demographic factors for each of the 14 harm types (p-values were corrected by multiplying with five). Table 3 shows the results for education context and Table 4 shows the results for job context. Below, we highlight a few significant differences across groups for each comparison factor, followed by explanations of possible socio-cultural issues that might have led to such differential perceptions.

4.5.1 ***Gender***. In the education context, the cross-gender differences in perceived harms show a consistent pattern: female-identifying participants expressed a higher level of concern about various harms for *all* data types compared to male-identifying participants, with many statistically significant differences, suggesting that these differences exist in the larger student population. This gender difference was most pronounced for personality traits, emotional states, and physical/cognitive impairment. Motivation and creativity had the fewest significant differences, which agrees with §4.5 that reported participants' comfort with using these data in both education and hiring contexts (see Figure 3).

These results align with prior research reporting that females were more concerned about privacy [90] and discrimination [88], as well as overall more negative toward AI decision-making [82], when compared to males. These findings might reflect participants' lived experiences in a society with many systemic and structural issues. Interestingly, female participants significantly differed from male participants about the harms from using problem-solving skills or impairment because these conditions are beyond their control. One could argue that learning and training activities improve those conditions; however, these facilities may not be equally accessible to males and females. Thus, this out-of-control perception may again be a result of the current societal structure.

In contrast, male and female participants did not differ in the hiring context (Table 4). This lack of a difference may have stemmed from more established norms or rules regarding the use of some of the data. For example, the use of demographics and disability status is strictly prohibited, while using other data types (such as problem-solving skills, personality, and motivation) may be acceptable (as a "business necessity") to data subjects.

4.5.2 ***Age***. In the education context (Table 3), older participants consistently perceived higher harms, with significant differences for with five out of six data types. These results validate prior findings that older adults are generally less appreciative of AI algorithms [82] and more concerned about privacy and fairness issues [90], possibly because of their less familiarity with technology. The only exception was demographic data, where both groups expressed a similar level of concern regarding their use across all harm categories. This result is not surprising, since the use of demographic data is explicitly prohibited by law, and harms arising from their use have been well-studied and widely known. Older participants also had heightened concerns for a larger number of harms when AI infers emotional state, motivation, or impairment. We speculate that these differences are reflective of being negatively judged, as society expects the academic journey to be 'linear': someone being a student after a certain age may be perceived as having low motivation, skills, or a physical or cognitive disability (including an inability to manage emotion).

We see the same pattern in the hiring context (Table 4): older participants perceived more harms regarding the use of demographics, personality, emotion, and disability, compared to younger ones. Note that the use of motivation and creativity in the hiring context was acceptable to the participants (see Figure 3), which did not change across age groups.

Comparing between the education and hiring contexts reveals further nuanced insights. For example, although (the average) perceived harms from inferring emotion or disability status were comparable between the education and hiring contexts (Figure 3), there were group-wise significant differences across the harm categories. In the education context, older participants were more concerned about after effects (e.g., harassment and manipulation) compared to younger participants; but in the hiring context, their concerns extended to the ethical implications and the working of the AI system (e.g., accuracy, bias, and inconsistency). One plausible explanation is that in the education domain, students with physical or cognitive disabilities are used to being offered accessibility assistance or counseling services; in contrast, such conditions may often lead to

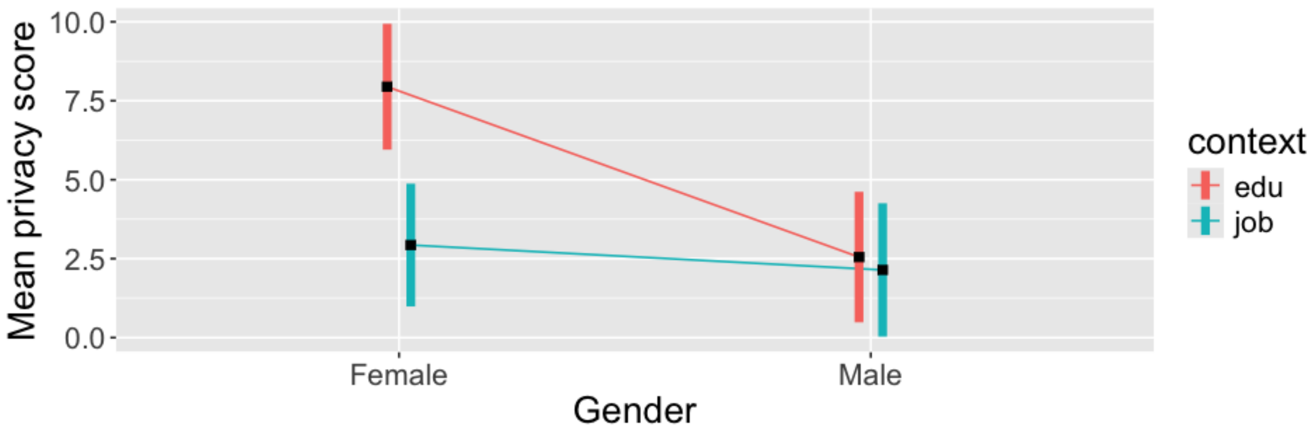


Figure 5: Mean and SE of perceived privacy harms across genders and contexts.

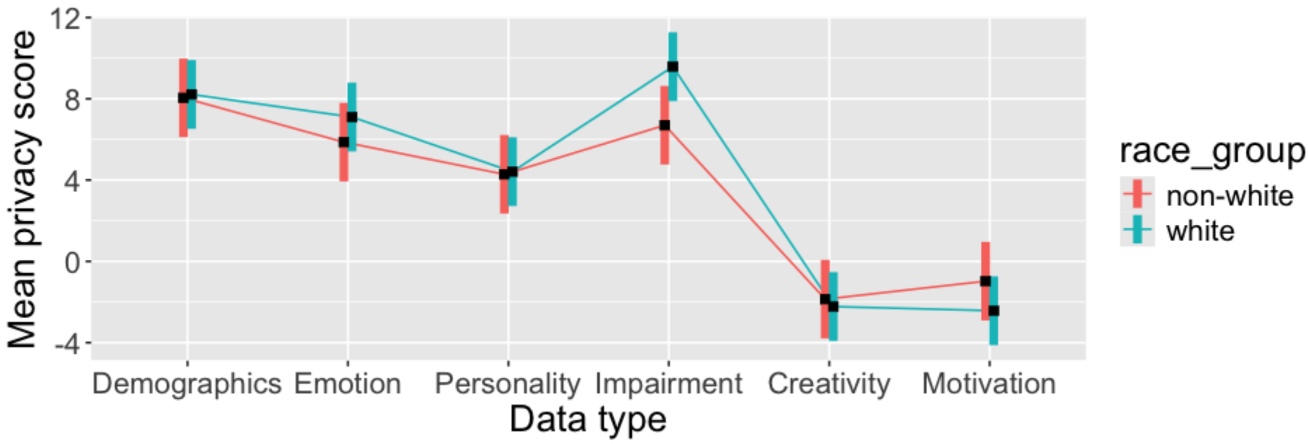


Figure 6: Mean and SE of perceived privacy harms across races and data types.

not getting hired [83]. Thus, older participants' harm perceptions may have been shaped by experiences of their own or surrounding people who went through recruitment processes.

4.5.3 ***Race.*** We found several interesting differences across races at the level of individual harms, despite the interactions involving race, data type, and context being non-significant for the composite score. In the education context (Table 3), white participants, compared to non-white participants, were more concerned about physical or cognitive disability inference due to manipulation, while the opposite was true for bias, stereotyping, and control. One reason could be that, while the prevalence of disability is roughly the same in white and non-white populations in the USA [3], the latter group historically has been experiencing more stereotyping and discrimination, and so harms from manipulation may stand out for the former group. Generally, participants were comfortable regarding the accuracy of inferring their motivation level and the reliability of it as a predictor (see negative means in Figure 3). The level of comfort, however, varied across races: Non-white participants were relatively more concerned regarding how accurately and reliably it can be inferred, possibly due to them having historically been marked as lazy and intellectually less capable than white people [24].

Likewise, in the hiring context, we found the most diverse harm perceptions across races (Table 4). First, a general trend: white participants expressed more concerns than non-white participants. Compared to non-white participants, white participants were more concerned about incorrect inferences of different data types (motivation, disability, creativity) and unreliability of the decision-making process because of their use. Looking at individual data types, we see that, for motivation and creativity, the use of which was generally acceptable, white participants were less comfortable with their use because of incorrect inference or unreliability, compared to non-white participants. For Emotion, we find opposite differences across races: white participants were more concerned about privacy, while non-white participants were more concerned about bias. This makes sense considering that studies have shown emotion detection models being biased against people of color, and being

[3] https://nces.ed.gov/fastfacts/display.asp?id=60

discriminated against is perhaps a bigger concern for marginalized populations than just the leaking of this (private) data, as privacy is often considered reserved for or afforded by only the privileged [72, 102]. We see additional insightful results for physical/cognitive disability: white participants perceived harms from privacy violations, defamation, and wrong inferences at a higher level than non-white participants, while the latter group were more concerned about harassment. This again possibly reflects the current social structure and lived experiences, and arguably being harassed (by others) does more tangible harm than feeling loss of privacy or reputation.

These results highlight the necessity to investigate at the individual harm levels, as opposing perceptions are lost when aggregated.

4.5.4 ***Education levels and disciplines***. In the education context (Table 3), comparing between undergraduate and graduate students, we found a significant difference in only one data type: undergraduates were more concerned about the use of motivation due to inaccurate predictions and biased decisions. Likewise, disciplinary difference was significant in one case; STEM majors, compared to non-STEM majors, were more concerned about losing autonomy when impairment was predicted.

The hiring context (Table 4) showed more differences across education levels: post-graduate students were consistently more concerned about diverse harms for several data types, most notably demographics, motivation, and disability. One possible explanation for these differences is that post-graduates are likely to have more experience in employment and interacting with automated hiring tools, and knowledge about appropriate and lawful hiring processes. Participants varied across study disciplines in one case only: non-STEM students expressed a higher level of concern about creativity being an unreliable predictor for hiring decisions.

Takeaways:

(1) Privacy harm perceptions across demographic factors are highly nuanced and diverse, possibly reflecting historical socio-economic and political factors, and participants' lived experiences.
(2) Surfacing diversity in perceived harms from a given technology can guide mitigating techniques, e.g., an algorithm deployed at a university primarily serving a specific demographic may prioritize mitigating harms to that population.

Table 3: Group-wise comparisons of harm perceptions in the education context using the Mann–Whitney U test. Cells show the z-statistics. Colored cells represent significant differences (after Bonferroni correction). Blue and orange backgrounds represent positive and negative z-statistics, respectively. *=p<.05, **=p<.01, ***=p<.001.

	Comparison group	Ethics	Fairness	Inaccuracy	Privacy	Consistency	Autonomy	Stereotyping	Reliability	Defamation	Harassment	Manipulation	Stress	Wrong inference	Control
Demographics	Gender	−1.46	-1.84^{*}	-2.28^{*}	-2.09^{*}	-3.14^{***}	−1.52	-2.42^{**}	-2.27^{*}	-2.58^{**}	−1.05	−0.39	−1.71	-2.69^{**}	−1.17
	Age	−0.29	−0.08	0.92	−1.13	−0.90	−0.29	−0.58	−0.05	−1.83	−1.08	−0.91	0.56	−0.37	0.06
	Edu level	0.08	0.00	−0.11	0.01	0.42	0.22	−0.40	−1.26	−0.09	0.24	0.12	0.47	0.45	0.56
	Race	0.65	1.45	1.08	−0.19	0.61	−0.09	0.70	−0.41	−0.20	1.07	0.20	−0.77	−0.84	1.10
	Major	−0.37	0.54	1.37	−0.60	0.85	1.28	1.12	−0.02	0.45	−0.20	−0.01	−0.03	−0.76	1.20
Personality	Gender	-2.18^{*}	-2.58^{**}	-3.23^{***}	-2.00^{*}	-2.19^{*}	-2.46^{*}	-3.36^{***}	-2.89^{**}	-2.69^{**}	−0.66	−1.78	-2.54^{**}	-2.87^{**}	−1.18
	Age	-2.48^{*}	−1.05	−1.11	−0.93	−0.91	−0.76	−0.79	−0.55	−1.60	-2.97^{**}	-1.97^{*}	−1.42	−1.05	−0.18
	Edu level	0.53	1.27	1.01	1.30	0.81	0.19	0.97	1.24	1.20	−0.73	0.70	−0.10	1.37	0.17
	Race	0.67	1.01	0.58	−0.73	−0.31	0.90	0.21	−0.11	0.58	0.42	0.11	1.07	0.91	−1.09
	Major	−0.37	0.53	1.83	0.81	0.66	1.26	1.78	−0.47	0.25	0.28	0.68	1.22	−0.08	1.53
Emotion	Gender	-2.36^{*}	-1.83^{*}	-2.68^{**}	-3.73^{***}	-2.19^{*}	-2.53^{**}	-2.54^{**}	-2.68^{**}	-2.34^{*}	−1.51	−1.53	-3.34^{***}	-2.34^{*}	−1.65
	Age	−0.70	-2.54^{**}	−1.49	−0.40	−0.75	−0.99	-1.94^{*}	−1.39	-2.82^{**}	-2.53^{**}	-2.51^{**}	−0.17	−0.99	−0.79
	Edu level	−0.63	−0.20	−0.99	−0.35	−0.32	−0.67	−1.76	0.69	−1.05	−1.29	−0.93	−0.96	0.43	−0.62
	Race	0.53	0.85	1.38	1.03	0.76	0.74	1.81	−0.83	0.29	−0.96	0.13	0.65	0.29	−0.71
	Major	0.57	1.19	1.22	0.46	0.01	0.97	0.42	0.10	−0.71	−0.30	0.70	0.31	0.24	1.52
Motivation	Gender	−1.26	−0.54	-2.37^{*}	−0.98	−1.81	-2.01^{*}	−1.61	−0.44	−1.42	−0.14	−0.23	−1.56	−1.12	−0.40
	Age	-2.76^{**}	-2.31^{*}	-2.87^{**}	-2.08^{*}	−0.92	−1.89	−1.64	−1.85	−1.69	-1.97^{*}	−1.57	-1.94^{*}	−1.13	-2.25^{*}
	Edu level	−0.43	-2.34^{*}	-2.07^{*}	−0.48	−0.85	−0.65	−1.06	−1.23	−1.43	−1.44	−1.07	−0.38	−0.66	−1.36
	Race	−1.50	1.05	0.29	-1.98^{*}	−1.22	−1.63	0.07	-2.32^{*}	−0.83	−1.06	−1.28	−0.28	-2.40^{*}	-2.84^{**}
	Major	−0.07	1.51	0.66	−0.65	0.12	−0.15	0.55	−0.64	−0.40	0.37	1.34	0.52	−0.22	−0.01
Creativity	Gender	−1.69	−1.66	-2.07^{*}	-2.37^{*}	-2.36^{*}	-2.32^{*}	−1.20	-1.96^{*}	−1.45	−0.47	−0.64	-2.66^{**}	-2.74^{**}	-2.08^{*}
	Age	−1.71	−1.71	−0.99	−1.18	-2.05^{*}	−0.75	−0.91	−0.67	−1.87	−1.33	-2.22^{*}	−1.01	−0.89	−0.77
	Edu level	−0.49	−0.30	0.12	1.13	0.40	0.48	−0.38	0.71	0.14	−0.18	0.68	−0.07	−0.28	0.19
	Race	−1.10	0.28	0.49	−0.25	0.89	−1.01	−0.45	0.13	−1.43	−1.08	−0.55	−0.34	1.17	−0.13
	Major	0.60	−0.22	0.25	1.18	−0.57	0.38	0.59	−0.04	−0.06	−0.08	−0.21	−0.24	−0.02	0.59
Disability	Gender	-2.95^{**}	-3.20^{***}	-3.20^{***}	-3.17^{**}	-2.70^{**}	-2.42^{*}	-2.66^{**}	-2.65^{**}	−1.73	-2.24^{*}	-2.35^{*}	-3.00^{**}	-3.68^{***}	-2.77^{**}
	Age	−0.46	−1.59	−1.63	−0.47	-2.78^{**}	−1.06	−0.91	-2.18^{*}	−1.87	-2.28^{*}	-2.68^{**}	−1.53	−1.66	−0.81
	Edu level	0.21	0.46	1.10	1.38	0.08	0.87	0.56	0.37	1.16	−0.48	0.81	0.42	0.50	0.10
	Race	1.69	2.70^{**}	1.06	0.85	1.30	0.09	2.77^{**}	−0.40	1.37	1.79	2.30^{*}	0.86	0.85	2.19^{*}
	Major	0.66	1.13	1.49	1.27	0.38	1.98^{*}	1.22	0.80	1.78	0.97	1.23	1.22	0.54	0.34

Table 4: Group-wise comparisons of harm perceptions in the employment context using the Mann–Whitney U test. Cells show the z-statistics. Colored cells represent significant differences (after Bonferroni correction). Blue and orange backgrounds represent positive and negative z-statistics, respectively. *=p<.05, **=p<.01, ***=p<.001.

	Comparison group	Ethics	Fairness	Inaccuracy	Privacy	Consistency	Autonomy	Stereotyping	Reliability	Defamation	Harassment	Manipulation	Stress	Wrong inference	Control
Demographics	Gender	0.02	0.30	0.71	−1.31	0.54	0.24	−1.36	−0.66	0.44	−0.21	0.89	0.04	−0.49	−0.14
	Age	−1.60	−0.86	−1.09	−0.87	−0.48	−1.24	-1.87^{*}	−0.68	−1.34	−0.12	−0.49	−0.49	−1.14	−0.27
	Edu level	−0.99	−0.72	−1.11	-1.94^{*}	−0.93	-2.51^{**}	−1.14	−1.87	−0.83	−1.71	-3.01^{**}	-2.34^{*}	−1.26	−0.29
	Race	0.22	0.60	0.64	0.30	0.44	−0.74	0.05	0.67	−0.66	0.48	−0.25	−0.37	0.72	−0.39
	Major	0.31	−1.18	−1.51	0.54	−0.50	−0.09	−0.30	−0.69	−0.55	−0.04	−0.15	−0.54	−0.59	−0.89
Personality	Gender	−0.60	−0.09	−0.07	−1.56	0.07	−0.73	−0.37	−0.29	0.33	−1.28	−0.24	−1.26	0.44	−1.57
	Age	-2.04^{*}	-1.88^{*}	−1.69	−0.60	−1.71	−1.58	−0.98	-2.08^{*}	−0.04	-1.97^{*}	−0.33	−0.39	−1.48	−0.79
	Edu level	−1.87	−1.78	−0.39	-2.92^{**}	−1.10	−1.61	−1.60	−0.75	−1.20	−0.32	−1.70	−0.93	−1.85	−0.85
	Race	1.05	−0.25	0.98	1.69	1.58	−0.50	−0.43	1.36	−0.46	-2.03^{*}	−0.47	0.78	2.05^{*}	−0.80
	Major	−0.01	−0.68	0.24	−1.24	−0.13	−0.28	0.90	0.08	−0.14	0.24	1.01	0.47	−0.49	−0.99
Emotion	Gender	−1.52	−1.19	−0.73	−0.66	0.05	−0.30	−1.63	−1.48	−0.05	−0.28	0.68	−1.25	−1.27	−1.76
	Age	-2.19^{*}	-3.01^{**}	-2.66^{**}	−1.49	-1.85^{*}	−1.52	-2.32^{*}	-2.45^{*}	-2.74^{**}	-2.35^{*}	−1.57	-1.94^{*}	-2.23^{*}	-2.14^{*}
	Edu level	−0.89	−1.21	−0.65	−0.60	−0.28	-1.99^{*}	−1.70	−0.40	−1.12	−0.13	−1.00	−0.08	−0.52	−1.41
	Race	1.42	2.02^{*}	1.75	2.04^{*}	0.99	1.05	0.99	1.55	0.17	−0.63	0.87	1.63	1.69	−0.11
	Major	−1.02	−0.83	−0.54	0.38	−0.18	−0.73	−0.28	−0.24	0.05	0.33	0.58	−0.54	−0.67	−0.34
Motivation	Gender	−0.16	−0.11	0.74	−0.89	−0.83	−0.68	−0.18	−0.01	−0.70	−1.14	−0.28	0.06	1.27	−1.26
	Age	−1.63	−1.09	−0.63	0.08	−0.77	0.01	−1.12	−1.45	−0.50	−1.56	−1.23	1.35	−1.22	0.01
	Edu level	-3.09^{**}	−1.89	−1.75	-1.89^{*}	−0.92	-2.07^{*}	−1.13	-2.04^{*}	−0.31	−0.60	−0.67	−0.51	−1.21	−1.51
	Race	1.02	0.71	1.36	−0.04	0.98	0.07	−0.85	2.22^{*}	−0.51	−0.91	−0.38	0.14	2.61^{**}	0.06
	Major	−0.58	−0.49	−0.15	−0.68	0.05	−0.13	0.92	−0.59	0.38	−0.26	0.28	0.00	−1.53	−0.96
Creativity	Gender	0.13	0.56	−0.77	−0.48	−1.75	−1.59	−0.34	−0.88	−0.04	0.29	0.78	0.10	0.60	−0.20
	Age	-2.52^{**}	-2.17^{*}	−1.70	−0.54	−0.80	-1.93^{*}	-2.17^{*}	−0.86	−1.60	−1.17	−1.81	−0.58	−1.52	−0.11
	Edu level	-1.97^{*}	−1.20	−0.25	−1.20	−0.24	−1.34	−1.47	−1.55	0.06	−0.14	−1.19	−0.68	−1.13	-2.11^{*}
	Race	0.87	0.71	0.41	0.00	−0.03	0.45	0.54	0.88	0.21	−0.13	0.58	−0.42	2.06^{*}	0.05
	Major	0.97	−0.00	−0.66	0.08	−0.89	−0.07	1.04	-2.06^{*}	0.33	0.71	1.09	0.28	−1.59	−0.28
Disability	Gender	−0.76	−1.13	0.32	−1.12	−0.51	−0.73	−0.99	−0.38	−0.56	0.43	0.10	−0.59	−0.47	−1.15
	Age	-2.43^{*}	-2.03^{*}	-2.64^{**}	−1.71	-3.02^{**}	−1.73	-2.71^{**}	−1.77	-2.00^{*}	−1.29	−1.68	−1.88	-3.03^{**}	-2.13^{*}
	Edu level	−1.70	−1.63	-2.24^{*}	−1.66	-2.26^{*}	−0.86	−0.96	−1.80	−1.43	−1.56	-1.90^{*}	−0.83	−1.90	-2.04^{*}
	Race	1.86	1.37	1.54	2.50^{**}	1.79	0.37	1.78	1.06	2.31^{*}	-1.97^{*}	0.34	1.40	2.92^{**}	0.63
	Major	−0.94	−0.48	0.12	0.28	−1.31	−0.71	0.30	−0.74	−1.32	−0.12	−0.95	−1.02	−1.29	0.10

5 Discussions and conclusions

In this paper, we proposed and operationalized a harm-centric conceptualization of privacy. To do that, we investigated perceived privacy harms in the context of using AI models in education and employment. Below, we discuss the theoretical and practical implications of this research.

Advancing the understanding of privacy. A general definition of privacy that captures all its aspects remains elusive [103]. Several conceptualizations have emerged over the decades; each of which has increasingly clarified our understanding of privacy and contributed to technical innovations in protecting it. Here, we propose a harm-centric view that conceptualizes privacy as a shield from different harms that can arise from personal data use. We demonstrated the reliability and validity of measuring privacy this way, suggesting that it can be used in other contexts to empirically understand privacy. However, one study does not establish general validity and reliability; any framework/theory requires multiple refinement and validation steps. This paper lays the groundwork of this framework, and future studies could validate it in other contexts (with different harms applicable to those contexts).

Nuanced discovery of privacy harms through the lens of procedural justice. Our approach surfaced privacy harms that may go undetected by other frameworks. We show that people anticipate privacy harms even when data is not considered inherently "private", nor there is a leak of data leak or an active adversary. Importantly, using the lens of procedural justice enabled us to surface privacy harms from otherwise justifiable, or even desirable systems. For example, one could argue that automation would expedite identifying at-risk students and providing targeted assistance, which is desirable; however, procedural justice requires that the operating procedure of that system does not harm the affected population. Indeed, our data shows that participants feared being discriminated against by the algorithms because of how they work (e.g., implicitly relying on protected attributes), even if the outcome could be useful.

Our approach surfaced and empirically validated harms that were not included in prior taxonomies of AI harms(e.g., [4, 19, 50, 56]). Our process-centric lens discovered harms (e.g., inconsistency) that arise due to how AI models make predictions based on training data. Finally, we provide a more nuanced view of harm perceptions by investigating variations across demographics. For example, while harms like bias and manipulation are commonly included in many taxonomies [4, 19, 50], we show that they are not perceived equally by everyone, and tie that differential perceptions to socio-demographic factors.

Equitable privacy with informed prioritization. Our results show significant variances across population segments in their perception of privacy harms. Not only did participants vary along the degree of harm, but different segments felt vulnerable to different harms in a given context. Thus, this approach allows us to identify the group likely to be most impacted, even if it's a minority. This also may provide clarity on how to prioritize among mitigating harms when needed; for example, when simultaneously satisfying different criteria like accuracy, privacy, and fairness may be computationally infeasible [18, 37].

Threat modeling. Understanding how personal data use harms data subjects can be useful in threat modeling for a system that uses such data. For example, Massey *et al.*used Solove's taxonomy [103] for mapping system vulnerabilities to privacy-violating actions [76]. Recent privacy threat modeling frameworks, such as LINDDUN and PANOPTIC, have started to go beyond threats from malicious users or active adversaries, and include harms to data subjects, both intentional and unintentional [57, 67]. There can be a direct mapping from privacy harms to the threats a system poses to its users; thus, studying privacy following the harm-centered approach can greatly accelerate user-centered threat modeling.

Recommendations to improve privacy. Empirical insights from this study could guide concrete policy and technical interventions to improve privacy. For example, we find that participants were more concerned when certain data (e.g., demographics) is used in the education context than in the employment context. As our literature review shows, these data are widely used both in research and commercial tools to predict many outcomes about students. Fortunately, institutional policies largely dictate how students' data can be used, both internally and by external entities such as technology providers [60]. Thus, institutes could implement preventive measures, such as prohibiting the repurposing of trained models (to infer personal data) by, e.g., extracting internal representation on input data. Additionally, both educational institutions and commercial entities can use technical mechanisms such as adversarial censoring to remove problematic personal data embedded in trained models [42]. Besides these technical measures, more human oversight of AI decision-making, particularly by trained personnel, can prevent harms or minimize negative consequences. AI models in use could be regularly audited, e.g., using carefully constructed data to see how different their decisions are from expert judgment. In any high-stake context, we argue that AI decision should not be the final decision.

Future human-centric privacy research. A long-term goal of human-centric privacy research has been to link people's privacy mental model to their privacy behaviors. This has proven to be difficult; in particular, we still lack an understanding of how people conceptualize privacy and related constructs and how to reliably measure them [21]. Contextual investigation of harm-centric conceptualization of privacy seems promising to shed light on when people may desire privacy protection. Future work could investigate how a desire for privacy influences behaviors that are protective of privacy. Future work could also investigate how harm perceptions influence technology use that may risk other people's privacy [43]. In particular, the education domain has seen a dramatic rise in the use of AI-based tools that are not institutionally acquired and possibly comes with greater risks [59, 124]; thus, understanding how educators perceive harms to students can shed light in their interdependent privacy mental model.

To conclude, we demonstrated that measuring perceived harms from personal data use can be a reliable and valid quantification of people's conceptualization of privacy as a shield against such harms. Thus, this approach can be used as a building block in future research approach to understand privacy in other domains.

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A Appendix

A.1 Survey questionnaire

A.1.1 Instructions. **Please read the following paragraph carefully before answering any questions.**

(**Education context**) Many schools now use artificial intelligence-based (AI) tools to support student success, which might expedite the process and enable evaluating student data at scale. AI tools can collect and analyze a lot of data from different sources in a much shorter time compared to human educators. For example, an AI tool may analyze your academic records, interactions with learning management systems, campus mobility data, as well as profiles and activities on external sites (e.g., GitHub or personal websites). Using all these data it can infer different criteria and based on which it decides whether you might be at risk of dropping out of a course.

Considering this situation, please answer the following questions about your opinion on AI tools predicting information about you.

(**Hiring context**) Artificial intelligence-based (AI) tools are being used for recruitment, which might enable evaluating job applications faster. AI tools can collect and analyze a lot of data from different sources in a much shorter time compared to human recruiters. For example, an AI tool may analyze your resume, information available online (e.g., from personal websites or Github profiles), recorded interviews, as well as data collected by different tools you used for studying (including learning management systems and other apps) to infer certain attributes about you. It can then use all these data to quantify different criteria and based on them it decides whether you should be employed.

Considering this situation, please answer the following questions about your opinion on AI recruitment tools predicting information about you.

A.1.2 Questions. After the general (context-specific) instruction, we explained the three attributes (shown below), after each of the explanations, we asked the participants to rate the following statements using a 5-point Likert scale ("Strongly agree" to "Strongly disagree"). Each participant only answered questions about one decision context: dropout risk or job fit assessment. The [Attribute] field was populated with the three attributes listed below.

(1) Use of [Attribute] to decide [dropout risk | job fit] is unethical.
(2) Use of [Attribute] to decide [dropout risk | job fit] can lead to biased decisions.
(3) Use of [Attribute] to decide [dropout risk | job fit] can lead to inaccurate decisions.
(4) Use of [Attribute] to decide [dropout risk | job fit] violates my privacy.
(5) Use of [Attribute] to decide [dropout risk | job fit] can lead to inconsistency in the decision-making process.
(6) Use of [Attribute] to decide [dropout risk | job fit] goes against my freedom to choose how I want to be judged.
(7) Use of [Attribute] to decide [dropout risk | job fit] can lead to stereotyping.
(8) [Attribute] is not a reliable predictor of [dropout risk | job fit].
(9) Use of [Attribute] to decide [dropout risk | job fit] can lead to defamation.
(10) Use of [Attribute] to decide [dropout risk | job fit] can lead to harassment.
(11) Use of [Attribute] to decide [dropout risk | job fit] can lead to manipulation.
(12) Use of [Attribute] to decide [dropout risk | job fit] will create stress for me.
(13) [Attribute] cannot be correctly inferred to use it to decide [dropout risk | job fit].
(14) Use of [Attribute] is inappropriate since I have no control over [Attribute].

A.1.3 Attribute descriptions.

- **Demographics:** Demographic information, such as age, gender, and race, can be predicted using machine learning algorithms based on various sources of information, including writing style, images/videos, and how users interact with online platforms.
- **Personality Traits:** Personality traits, such as extroversion, conscientiousness, and openness, can be predicted using machine learning algorithms based on various sources of information, including social media activity, language use in communications, Video Interviews and Github Issues and Commits Activity.
- **Emotional State:** Emotional state, such as stress levels and emotional stability, can be predicted using machine learning algorithms based on various sources of information, including facial expressions in videos, tone of voice in audio recordings, and language use in written communications.
- **Motivation:** Motivation, in educational and employment contexts varies by individual, which makes it highly subjective, hence challenging to predict using machine learning algorithms.
- **Creativity and Problem Solving:** Creativity and problem solving, similarly to motivation, are used in hiring decisions and educational evaluations.
- **Physical and Cognitive Impairment:** Physical and Cognitive Impairment, such as physical and learning disabilities, can be predicted using machine learning algorithms through audio, image and video data.

A.1.4 Demographic questions. Finally, we asked questions about gender, age, current education level, and race.

A.2 Additional results

Table 5: Cronbach's Alpha (leave-one-out) and Item-Total Correlation (in parentheses) by Attribute for Employment Context

Item Removed	Demographics	Personality	Motivation	Emotion	Creativity	Impairment
-	0.94	0.92	0.93	0.93	0.94	0.95
unethical	0.93 (0.81)	0.91 (0.72)	0.93 (0.70)	0.93 (0.79)	0.94 (0.78)	0.95 (0.78)
biased	0.93 (0.70)	0.91 (0.70)	0.93 (0.70)	0.93 (0.68)	0.94 (0.74)	0.95 (0.74)
inaccurate	0.93 (0.76)	0.91 (0.69)	0.93 (0.70)	0.93 (0.73)	0.94 (0.75)	0.95 (0.77)
privacy	0.93 (0.64)	0.92 (0.65)	0.93 (0.70)	0.93 (0.70)	0.94 (0.69)	0.95 (0.78)
inconsistency	0.93 (0.68)	0.92 (0.65)	0.93 (0.71)	0.93 (0.73)	0.94 (0.76)	0.95 (0.79)
freedom	0.93 (0.68)	0.92 (0.64)	0.93 (0.70)	0.93 (0.69)	0.94 (0.73)	0.95 (0.78)
stereotyping	0.93 (0.71)	0.92 (0.64)	0.93 (0.70)	0.93 (0.70)	0.94 (0.74)	0.95 (0.71)
reliable	0.93 (0.70)	0.92 (0.61)	0.93 (0.68)	0.93 (0.62)	0.94 (0.70)	0.95 (0.74)
defamation	0.94 (0.61)	0.92 (0.64)	0.93 (0.66)	0.93 (0.68)	0.94 (0.71)	0.95 (0.69)
harassment	0.93 (0.72)	0.92 (0.58)	0.93 (0.62)	0.93 (0.65)	0.94 (0.68)	0.95 (0.73)
manipulation	0.93 (0.66)	0.92 (0.66)	0.93 (0.68)	0.93 (0.71)	0.94 (0.77)	0.95 (0.68)
stress	0.93 (0.67)	0.92 (0.63)	0.93 (0.67)	0.93 (0.70)	0.94 (0.70)	0.95 (0.69)
inference	0.93 (0.72)	0.92 (0.64)	0.93 (0.67)	0.93 (0.60)	0.94 (0.69)	0.95 (0.72)
control	0.93 (0.69)	0.92 (0.60)	0.93 (0.59)	0.93 (0.61)	0.94 (0.60)	0.95 (0.74)

Table 6: Number of participants when grouped by demographic factors

Grouping	Education context	Employment context
White	112	107
Non-White	75	82
STEM	90	99
Non-STEM	110	101
Male	89	84
Female	97	102
Undergrad	96	105
Post-grad	86	83
18–24 years old	79	82
>24 years old	110	107

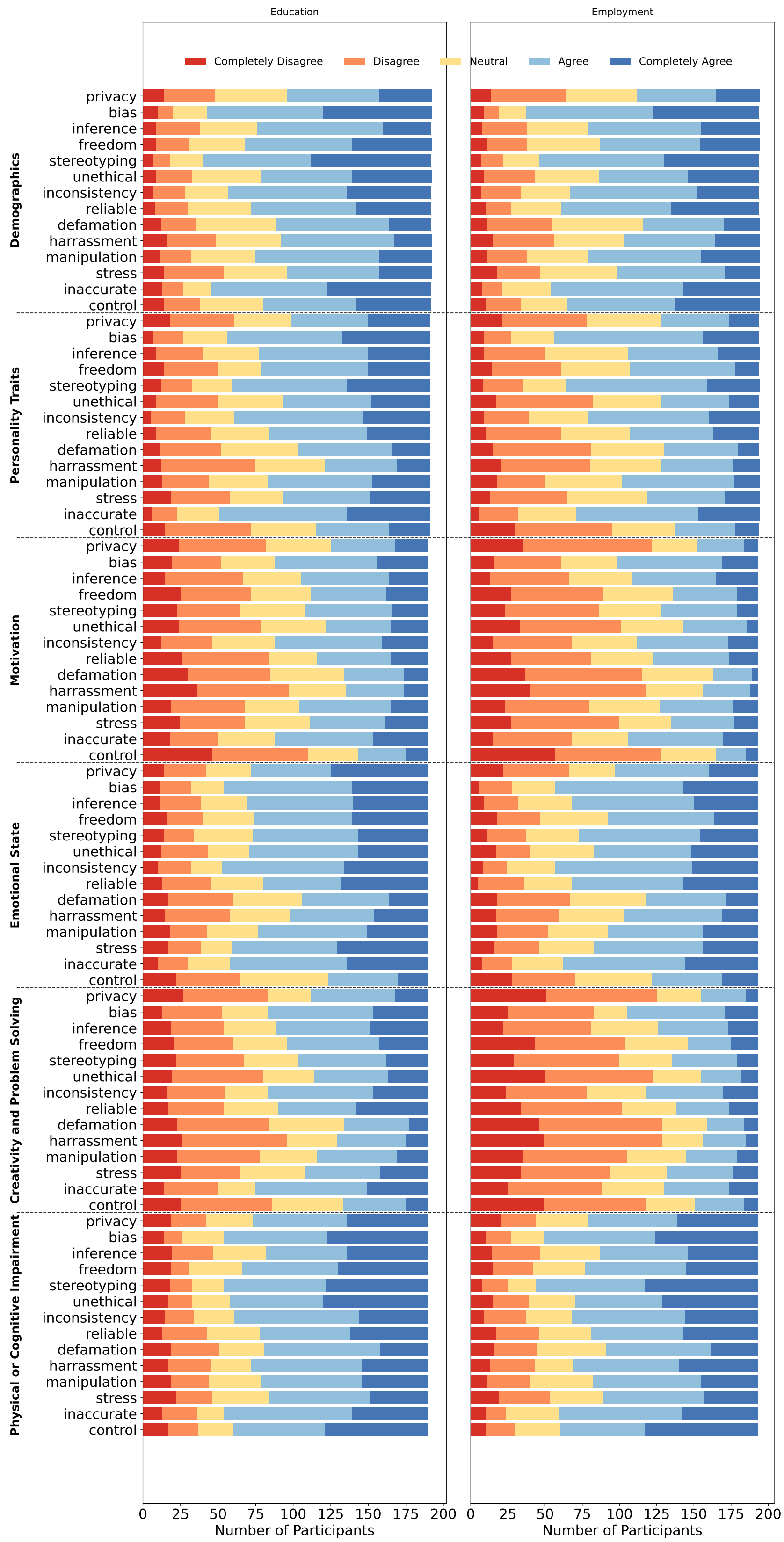

Figure 7: Distribution of participants' responses to harm statements (left: education, right: employment).